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A three-variety automatic and non-intrusive computer vision system for the estimation of orange fruit pH value

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ABSTRACT

An automatic 3-variety computer vision orange fruit pH value assessment system in the visible-range is presented, including each 100 different color images from Bam, Blood and Thomson orange of which the true pH has been measured and is known in advance. A total of 452 features are extracted from segmented orange color images. Results with repeated trials include: true versus estimated mean pH values, true minus estimated pH values boxplots, fitness regression dispersion plots and various error measure boxplots, for both single orange variety as well as the three-orange varieties altogether (test set), showing consistent results over all three orange varieties. Regression coefficient for pH estimation in Bam, Blood and Thomson orange varieties, were 0.950, 0.935 and 0.957, respectively. Results show that the hybrid ANN-ABC estimates pH values in orange quite similarly among orange varieties, implying that properly pH values estimation is possible for different orange varieties, regardless of orange type.

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1. Introduction

Non-destructive estimation methods are very useful in packaging systems, since based on them the possibility to the online prediction of inner properties of fruits can be made. Machine vision is a technique of Electronics field which is useful to be applied in modern pH value estimation systems. Regarding the relationship between electronic systems based on machine vision, it should be mentioned that there are two main types of intelligence: either artificial intelligence or machine intelligence. In machine vision, which is useful for developing electronic pH value estimation systems, mainly machine intelligence has been used. These systems could be either hardware or software based. We focus on software-based machine vision systems here. The number of recent techniques and applications behind the field of computer vision is very broad and huge, having among the first ones (theoretical) the level sets [17], spectral clustering and cellular learning automata [7], to name a few, and among the second (applications) texture recognition [5], lung nodule detection [10,19], real-time pedestrian detection [11,22], iris recognition [28], and coastline detection [16], to name a few.

Regarding the particular topic of interest here, different researchers have recently proposed new programs to determine the values of inner properties of various fruits. Generally speaking, the accuracy of such systems is not very high. Based on the view point of some consumers, the main quality aspects for fruits and vegetables are size, appearance and color, but other persons believe that inner properties of fruits and vegetables such as titratable acidity (TA), soluble solids content (SSC), soluble solids to acid (SSC/TA) ratio and texture, are more important than

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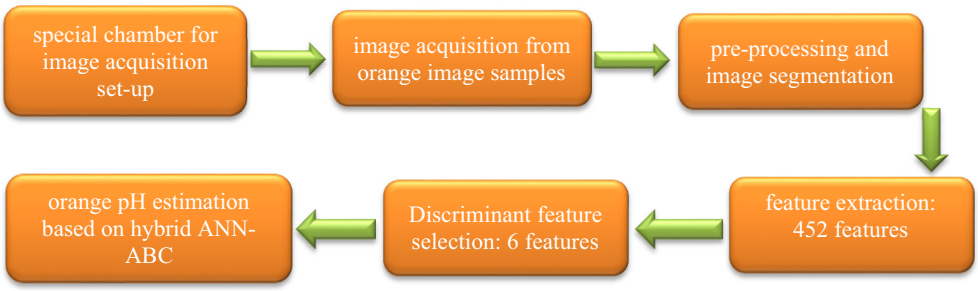


Fig. 1. Different stages in the design of a non-destructive visible-range computer vision system for the estimation of the pH value in orange fruit.

Table 1
Mean and std. of several physical properties and pH values measured over Bam, Blood and Thomson variety oranges.

Orange variety	Length (mm)	Width (mm)	Height (mm)	Weight (g)	Volume (ml)	pH value
Bam	63.01 ± 3.69	57.47 ± 3.08	62.61 ± 3.55	116.97 ± 16.73	120.53 ± 19.39	3.76 ± 0.52
Blood	58.77 ± 2.85	56.35 ± 3.28	58.23 ± 2.87	102.75 ± 11.31	111.86 ± 13.39	3.80 ± 0.41
Thomson	71.02 ± 4.33	70.56 ± 4.86	70.16 ± 4.37	193.64 ± 95.58	198.82 ± 30.71	4.68 ± 0.67

Note: 100 samples of each orange variety were used, totaling 300 orange samples. *Devices used in measurements*: Guanglu digital caliper model HB-102-111, China, with an accuracy of 0.01 mm; electronic balance A&D, model AND GF3000, Japan, with a precision of ±0.01 g; pH-meter PH/ORP/ METER model S.N0901020, Taiwan. *Volume measurement procedure*: each orange was submerged in a container full of water, and the weight (W) of the displaced water was measured in grams (g), resulting in orange volume (cm^3) of $V = W/\rho$, being ρ water density (g/cm^3), typically 1 (g/cm^3).

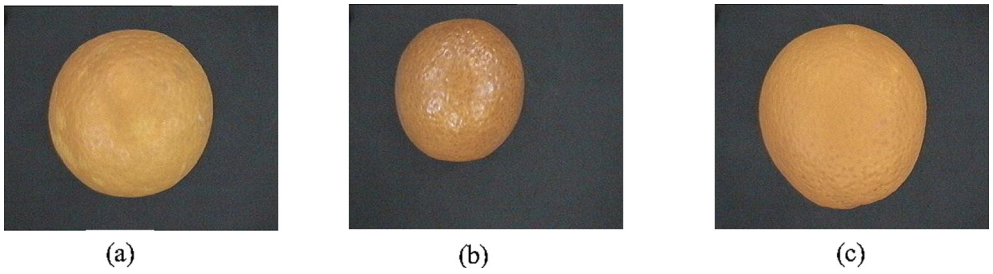


Fig. 2. A sample image of each orange variety. (a): Bam, (b): Blood, (c): Thomson.



Fig. 3. Special purpose hand-made chamber for image acquisition.

external properties or appearance [1,21,3,4,13]. Several studies for grading based on size, mass, or volume are already available in the literature, like for instance [18,6,26,24]. The common basics under all these methods is image processing. In fact, only after proper image segmentation and feature extraction, based on different linear or non-linear methods, various inner fruit properties are estimated. Some valid examples of researches that focus over the determination of inner properties in a non-destructive fashion, include [27,2,23,25].

Rungpichayapichet et al. [23] proposed a non-destructive method for predicting physicochemical characteristics of firmness, TSS, and TA of mango fruit at different stages of ripening using hyperspectral images. For the training phase of the proposed algorithm, 160 mango fruit samples were used, unhealthy mango fruits were isolated by visual inspection, and the remainder mangos were selected for testing purposes. Predictive models were developed using spectral data in visible and infrared regions by means of a least partial squares regression. A multiple linear regression method was used to identify 16 important factors in firmness, 17 factors in TA, and 20 factors in TSS. Results showed that there is an important correlation between hyperspectral images and physicochemical properties of firmness, TSS and TA. The value of the determination coefficient and the root Mean Square Error (RMSE) between hyperspectral images and firmness, R^2 = 0.81 and RMSE = 2.83, hyperspectral and TSS R^2 = 0.81 and RMSE = 0.24% and hyperspectral and TA R^2 = 0.5 and RMSE = 2.0%.

Munera et al. [20] used visible and near-infrared spectral reflection images to determine the physicochemical properties and evaluate internal quality of two nectarine varieties, BigTop and Magique, during the ripeness period using both ripening index (RPI) and internal quality index (IQI). Fruits were collected from

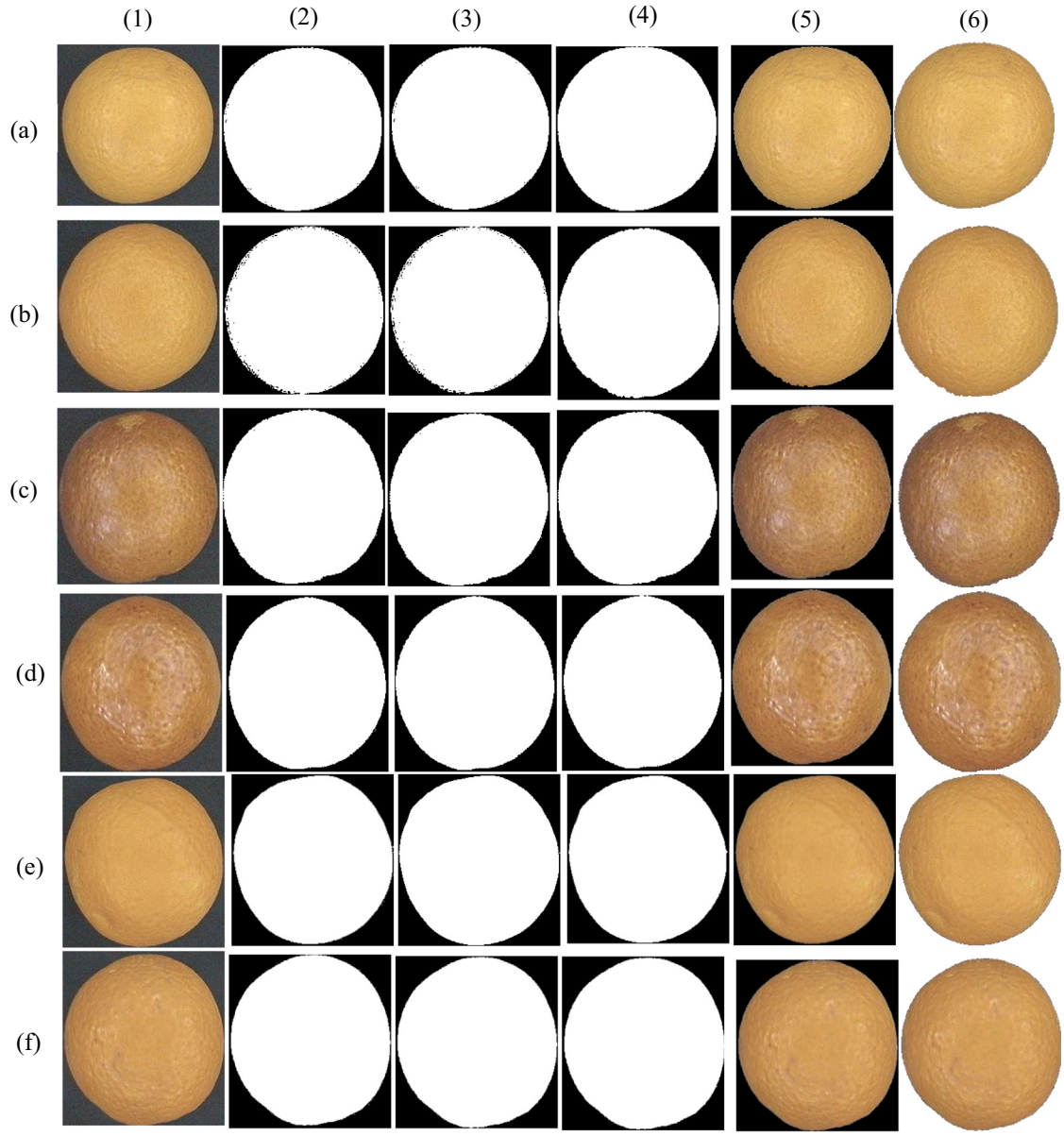


Fig. 4. Six different stages of image segmentation. (rows a and b): Bam orange, (rows c and d): Blood orange, (rows e and f): Thomson orange. (column 1): original orange image, (column 2): binary image, (column 3): binary image after the first stage of refinement, (column 4): binary image after closing operation, (column 5): segmentation with black background, (column 6): final segmented image without background.

Table 2
Mean and standard deviation (112) features for 16 different color spaces: 16 × 7 = 112 features.

Color space	Extracted features from each color space (mean and std.)						
RGB, HSV, YIQ, YCbCr, CMY, HIS, L*a*b*, JPEG-YCbCr, YDbDr, YPbPr, YUV, HSL, XYZ, Luv, LCH, and CAT02 LMS	First component mean	Second component mean	Third component mean	Three components mean together	Standard deviation of the first component	Standard deviation of the second component	Standard deviation of the third component

garden, in Lerida of Spain, during the maturity period. Samples were placed in groups with 25 samples and stored under controlled conditions (temperature 15 °C and relative humidity 90%). In each experiment, in addition to the extraction of properties like firmness, TSS, TA and meat color, using an industrial camera (CoolSNAP ES, Photometrics, AZ, USA), 60 hyperspectral pictures were taken, in the range of 450 nm to 1040 nm wavelengths (every 10 nm, one image). In the next stage, RPI and IQI indeces were computed in order to link spectral information with physicochemical properties from nectarine, and input in a least partial squares regression system. Results showed a highest correlation coefficient of determination value of 0.87.

Table 3
Vegetation indeces (224) color feature definitions: name and formal mathematical definition (16 × 14 = 224 features): RGB color space.

Extracted features	The formula for calculating the extracted features
The normalized first component of RGB color space	$R_n = R/(R+G+B)$
The normalized second component of RGB color space	$G_n = G/(R+G+B)$
The normalized third component of RGB color space	$B_n = B/(R+G+B)$
Gray channel	$luminance = 0.2898 \times R_n + 0.5870 \times Gn + 0.1140 \times B_n$
Additional green [29]	$EXG = 2 \times G_n - R_n - B_n$
Additional red [14]	$EXR = 1.4 \times R_n - G_n$
Color index for extracted vegetation cover [12]	$CIVE = 0.441 \times R_n - 0.811 \times G_n + 0.385 \times B_n + 18.78$
The subtraction between additional green and additional red [15]	$EXGR = EXG - EXR$
Normalized difference index [30]	$NDI = (G_n - B_n)/(G_n + B_n)$
Green index minus blue [29]	$GB = (G_n - B_n)$
Red-blue contrast [8]	$RBI = (R_n - B_n)/(R_n + B_n)$
Green-red index [8]	$ERI = (R_n - G_n) \times (R_n - B_n)$
Additional green index [8]	$EGI = (G_n - R_n) \times (G_n - B_n)$
Additional blue index [8]	$EBI = (B_n - G_n) \times (B_n - R_n)$

Note: despite the definition of vegetation indeces is given only for RGB color space, it can be extended to other 16 different color spaces used here, regardless if color space component has luminance or chrominance information inside. *Color spaces*: RGB, HSV, YIQ, YCbCr, CMY, HSI, L*a*b*, JPEG-YCbCr, YDbDr, YPbPr, YUV, HSL, XYZ, Luv, LCH and CAT02- LMS (c.f. Table 2).

Table 4
Texture features (80) extracted based on the gray level co-occurrence matrix: name.

Feature number	Feature name	Feature number	Feature name
1	Maximum probability	11	Inverse difference normalized (INN)
2	Sum of squares	12	Inverse difference moment normalized
3	Sum average	13	Diagonal moment
4	Sum variance	14	Second diagonal moment
5	Sum entropy	15	Variance
6	Difference variance	16	Mean
7	Difference entropy	17	Standard deviation
8	Information measure of correlation1	18	Coefficient of variation
9	Information measure of correlation2	19	Contrast
10	Inverse difference (INV) in homogeneity	20	Correlation

Note: a total of 20 (features) × 4 (angular orientations) = 80 texture features were computed. Angular orientations: 0°, 45°, 90° and 135°.

Table 5
Extracted shape features (20) from object: name.

Feature number	Feature name
1	Length
2	Width
3	Area
4	Logarithm of the ratio of length to width
5	The ratio of perimeter to broadness of object
6	The ratio of width to length
7	The ratio of area to length
8	Eccentricity
9	Perimeter
10	Orientation
11	ConvexArea
12	FilledArea
13	EquivDiameter
14	EulerNumber
15	Solidity
16	Elongation
17	Compactness
18	Extent
19	Aspect Ratio
20	The ratio of length to Perimeter

The aim of this study is to introduce a machine vision system for non-destructive determination of the pH values in three different orange varieties, Bam, Blood and Thomson, based on image processing and various hybrid Multilayer Perceptron (MLP) neural networks. This system may be useful in the packaging industry including grading based on the pH value. Fig. 1 shows the different stages in designing a computer vision system in order to estimate orange pH value in a non-destructive fashion.

2. Material and methods

2.1. Data collection

In this study, three different varieties of orange were used, including Bam, Blood and Thomson. A total of 100 samples of each variety, with physical mean properties as shown in Table 1, were randomly selected. Fig. 2 shows one sample of each variety for illustrative purposes. All experiments were performed in the Physical Laboratory of the Faculty of Agriculture Engineering, University of Razi in Kermanshah, Iran, using a pH meter (PH/ORP/Temperature meter, Taiwan), the pH value of all samples was measured. Immediately after measurement of true pH values, images were taken using a special purpose hand-made photographic chamber (Fig. 3). This chamber has several components including, digital camera (BOSCH, Portugal), frame grabber (Pinnacle, China), various lamp types (LED, Fluorescent and Tungsten), a dimmer for light intensity modulation and a PC. An off-the-shelf PC with an Intel Core i3 CFI processor, 330 M at 2.13 GHz, 4 GB of RAM and Windows 10 equipped with MatLab (ver. R2014a) and Microsoft Excel (ver. 2013) programs, was used. Color images were taken at a 352 × 288 pixels resolution. Camera was placed in the center of chamber. The distance between camera and samples was adjusted at a fix height value of 10 cm. For imaging purposes, white LEDs with an intensity of 50.33 lx. were selected.

2.2. Segmentation

Segmentation is a main step in machine vision systems. Based on a proper segmentation, the main object separates from the

Table 6
Values of MLP neural network parameters for the hidden layer.

Feature extraction method	Number of neuron	Number of layer	Transfer function	Backpropagation network training function	Backpropagation weight/bias learning function
Hybrid ANN-PSO	First layer: 10	1	First layer: tansig	trainlm	learnhd

Table 7
Selected discriminant features (6) using ANN-PSO approach for different orange varieties: Bam, Blood, Thomson and 3-orange varieties together.

Orange variety	Selected features
Bam	Normalized difference index of YIQ color space, Color index for extracted vegetation cover of YIQ color space, Additional second channel index of LCH color space, Additional second channel index of YPbPr color space, and Sum of squares for 90 degrees angle.
Blood	Normalized first component of RGB color space, Additional first channel index of HSV color space, Three components mean together of YCbCr color space, Dissimilarity for 90 degrees angle, Three components mean together of CMY color space, and Additional second channel index of YPbPr color space.
Thomson	Normalized first component of LCH color space, Autocorrelation for 90 degrees angle, The normalized third component of HSV color space, First-third contrast of LCH color space, First-third contrast of LAB color space, and Normalized first component of HSI color space.
3-varieties together	Mean for angle 90 degrees, Homogeneity, Inverse difference (INV) in homogeneity for 0 degrees angle, Information measure of correlation for 0 degrees angle, Three components altogether mean of LCH color space, and Sum of squares for 0 degrees angle.

Table 8
Possible range of values for five different parameters of the ANN.

Parameter name	Possible parameter values
Number of layers	{1,2,3}
Number of neurons in each layer	{0,1,2,3, . . ., 25}
Transfer function	$Tansig_1$ $logsig_2$ $purelin_3$ $hardlim_4$ $compet_5$ $hardlims_6$ $netinv_7$ $poslin_8$ $radbas_9$ $satlin_{10}$ $satlins_{11}$ $softmax_{12}$ $tribas_{13}$
Back-propagation network training function	$trainlm_1$ $trainbfg_2$ $trainrp_3$ $traingd_4$ $traincgf_5$ $traincgp_6$ $traincgb_7$ $trainscg_8$ $trainoss_9$ $traingda_{10}$ $traingdx_{11}$ $trainb_{12}$ $trainbfgc_{13}$ $trainbr_{14}$ $trainbuwb_{15}$ $trainc_{16}$ $traingdm_{17}$ $trainr_{18}$ $trains_{19}$
Back-propagation weight/bias learning function	$learngdm_1$ $learngd_2$ $learncon_3$ $learnh_4$ $learnhd_5$ $learnis_6$ $learnk_7$ $learnlv1_8$ $learnlv2_9$ $learnos_{10}$ $learnp_{11}$ $learnpn_{12}$ $learnsom_{13}$ $learnsomb_{14}$ $learnwh_{15}$

background. In this study, a threshold based on the first component of RGB color space was used for segmentation purposes. Eq. (1) defines this segmentation threshold:

$$R(i,j) < 120 \quad (1)$$

Based on this threshold, if the first component of RGB color space value is less than 120 that pixel is considered to belong to the background and omitted; otherwise it is considered as foreground orange image pixel and kept. After applying Eq. (1) some noise was apparent in segmented images. After conversion of image to binary image, all objects with a total number of pixels less than 100 were removed. Later on, we used a closing filter (*imclose* in MatLab) as a second stage of segmentation refinement. Fig. 4 shows the summarized different stages of segmentation process including two orange samples belonging to each orange variety.

2.3. Feature extraction

Feature extraction or feature computation, is one of the most important stage in order to properly train a Computer Vision system. Generally speaking, feature extraction plays a key role in Computer Vision because a larger numbers of extracted features increases the possibility of finding discriminant features with high separation capability of existing classes. For this reason, after a segmentation stage, different features were extracted from each orange image. All features were extracted in MatLab (ver. R2014a) using different mathematical formulae and methods.

In this study, a total number of 452 features of 5 different types: color (336), texture (80), histogram (6), moments (10) and shape (20), were extracted from each orange color image, as described next.

2.3.1. Color features

A total of 16 different color spaces were included as listed next: RGB, HSV, YIQ, YCbCr, CMY, HSI, L*a*b*, JPEG-YCbCr, YDbDr, YPbPr, YUV, HSL, XYZ, Luv, LCH and CAT02- LMS. Extracted features based on color information are shown in Table 2, totaling 16 (color spaces) × 7 (features) = 112 features. Table 3 shows additional color features (vegetation indeces) totalling 16 (color spaces) × 14 (features) = 224 features. Thus, the total number of color features over the 16 different color spaces was 336.

Table 9
Optimal values for the NN parameters determined using the ABC optimization algorithm.

Feature extraction method	Number neurons	Number layers	Transfer function	Backpropagation network training function	Backpropagation weight/bias learning function
Hybrid ANN-ABC	First layer: 13, second layer: 20, third layer: 14	3	First layer: poslin, second layer: poslin third layer: satlin	trainlm	learnh

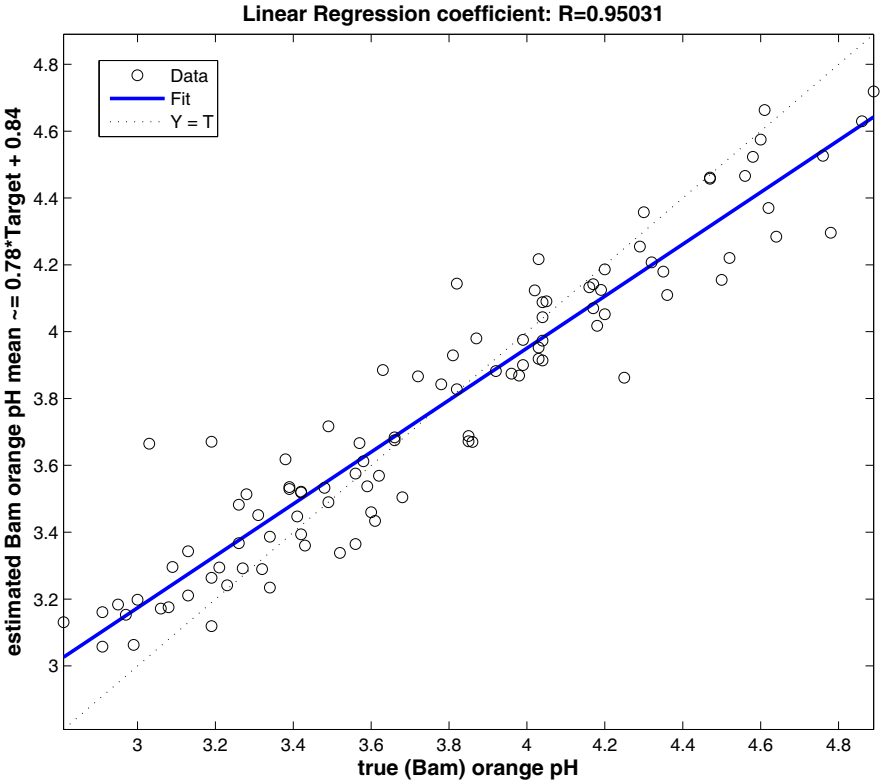


Fig. 5. Linear regression dispersion plot for Bam orange pH value (test set): mean estimated versus true (measured) orange fruit pH value. The linear best fit is also depicted in blue color line. 1000 uniform test sets where used to compute mean estimated pH values for each of the 100 oranges (depicted as a circular point in figure). On average, each orange is used 1000 repetitions *30 samples in test set/100 input oranges = 300 times, inside the test set. **Note**: for comparison purposes, the X = Y straight line is also depicted as a dotted line in figure.

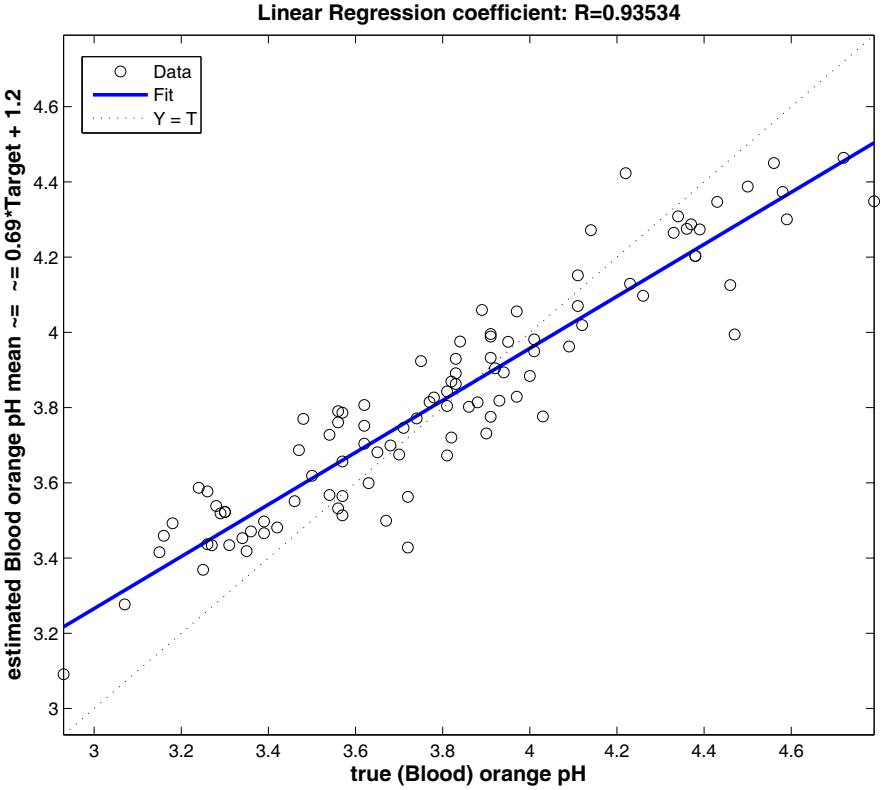


Fig. 6. Linear regression dispersion plot for Blood orange pH value (test set): mean estimated versus true (measured) orange fruit pH value. The linear best fit is also depicted in blue color line. 1000 uniform test sets where used to compute mean estimated pH values for each of the 100 oranges (depicted as a circular point in figure). On average, each orange is used 1000 repetitions *30 samples in test set/100 input oranges = 300 times, inside the test set. **Note**: for comparison purposes, the X = Y straight line is also depicted as a dotted line in figure.

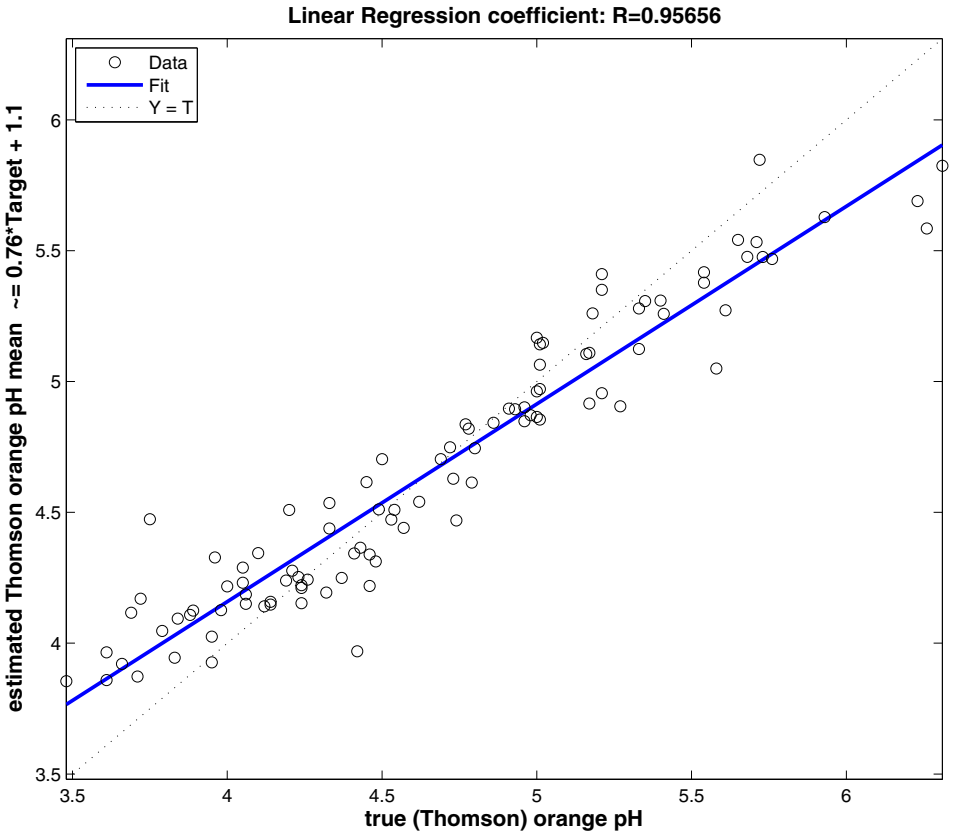


Fig. 7. Linear regression dispersion plot for Thomson orange pH value (test set): mean estimated versus true (measured) orange fruit pH value. The linear best fit is also depicted in blue color line. 1000 uniform test sets where used to compute mean estimated pH values for each of the 100 oranges (depicted as a circular point in figure). On average, each orange is used 1000 repetitions *30 samples in test set/100 input oranges = 300 times, inside the test set. **Note**: for comparison purposes, the X = Y straight line is also depicted as a dotted line in figure.

2.3.2. Texture features based on the gray level co-occurrence matrix

Pixel neighborhood is defined for different angles (0°, 45°, 90° and 135°) at a distance of 1 pixel. Table 4 shows 20 texture features extracted based on the gray level co-occurrence matrix. Total number of extracted texture features are 4 (angles) × 20 (features) = 80.

2.3.3. Histogram-based texture descriptors

Six histogram features were computed, which include: Average, Standard deviation, Homogeneity, Entropy, Smoothness and Third moment [9]; previous histogram features were extracted from each orange sample image.

2.3.4. Moment invariants

A total number of 10 computed moment features include: First to Seventh moment invariant, the Difference between First and Seventh moment invariant, Difference between Second and Sixth moment invariant and Difference between Third and Fifth moment invariant [9].

2.3.5. Shape features

Table 5 shows the 20 shape features that were extracted from each orange fruit image [9].

2.4. Suboptimal feature selection

The use of a total of 452 color, texture, histogram, moments and shape extracted features, described above, as input to the proposed method to the non-invasive estimation of the orange pH value is not adequate for two main reasons: the speed of the estimation of pH decreases as the number of inputs increases; and the accuracy of estimation of pH can also be reduced because of the existence of dependencies between features avoiding proper learning generalization capabilities. Therefore, it is essential to select the most effective (discriminant) features among all available ones; this way, the time required to compute them is also decreased considerably. Despite feature selection is not an image processing technique, it is mandatory. In fact, after the previous computation of the different features, feature selection was performed based on both statistical and non-statistical methods such as various hybrid neural networks approaches.

In this study, hybrid Artificial Neural Network-Particle Swarm Optimization (ANN-PSO) was used for suboptimal effective discriminant feature selection among the total 452 different extracted features. In this method, first all features are considered as one vector. In next step, different vectors with different sizes were selected based on the PSO optimization algorithm and transferred as inputs to an MLP neural network architecture. The output of MLP was the estimated orange pH regression values. We divided all input data into 3 disjoint sets: 70% training set, 15% validation set and remainder 15% test set, using at the same time uniform random sample selection (randomized trials) repeated an enough number of times in order to reach statistical significant results. MSE values of each vector were registered. Each vector that had the lowest MSE value was selected as an effective subset of features. Table 6 shows the optimal values of the MLP neural network parameters for the hidden layer. Table 7 shows discriminant selected features

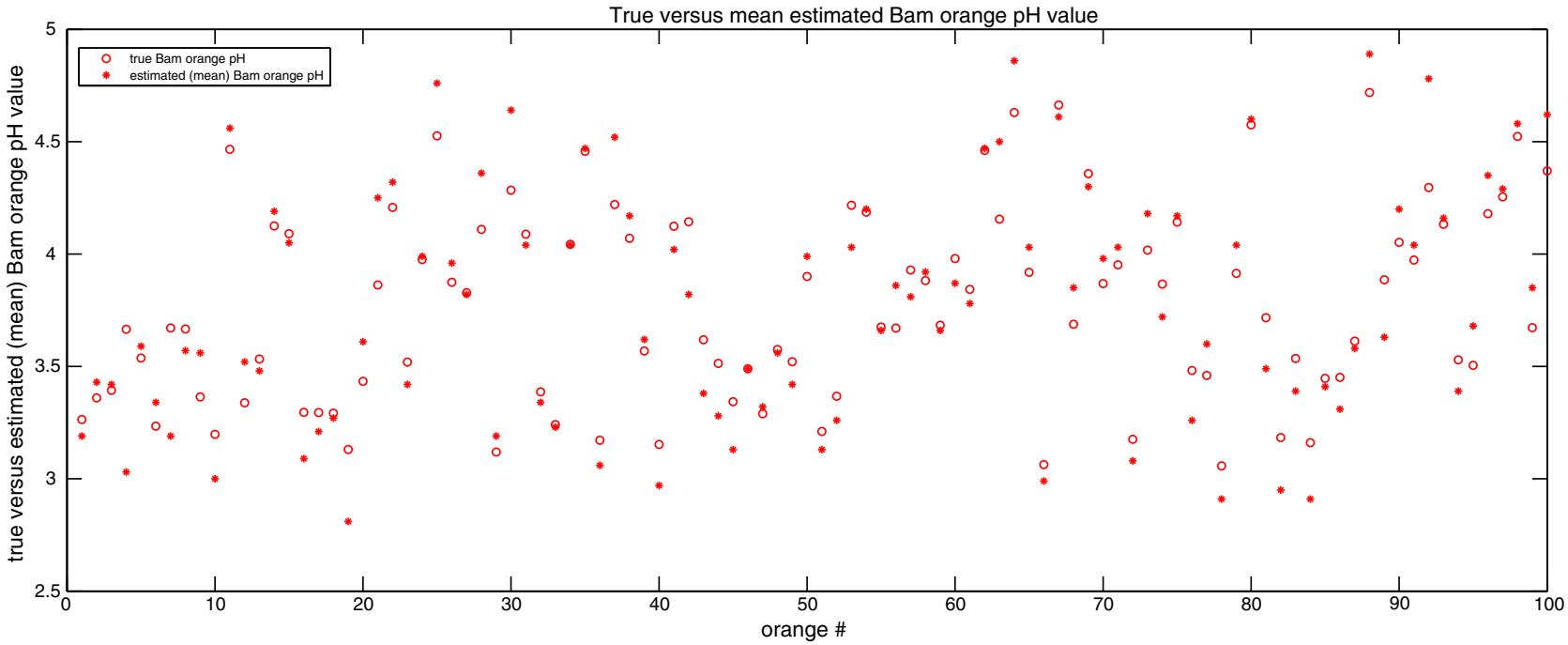


Fig. 8. True (red circle) versus mean estimated (red asterisk) Bam orange pH values (test set): 1000 uniform random test set samples simulations averaged; each orange # is used 300 times inside the test set, on average. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

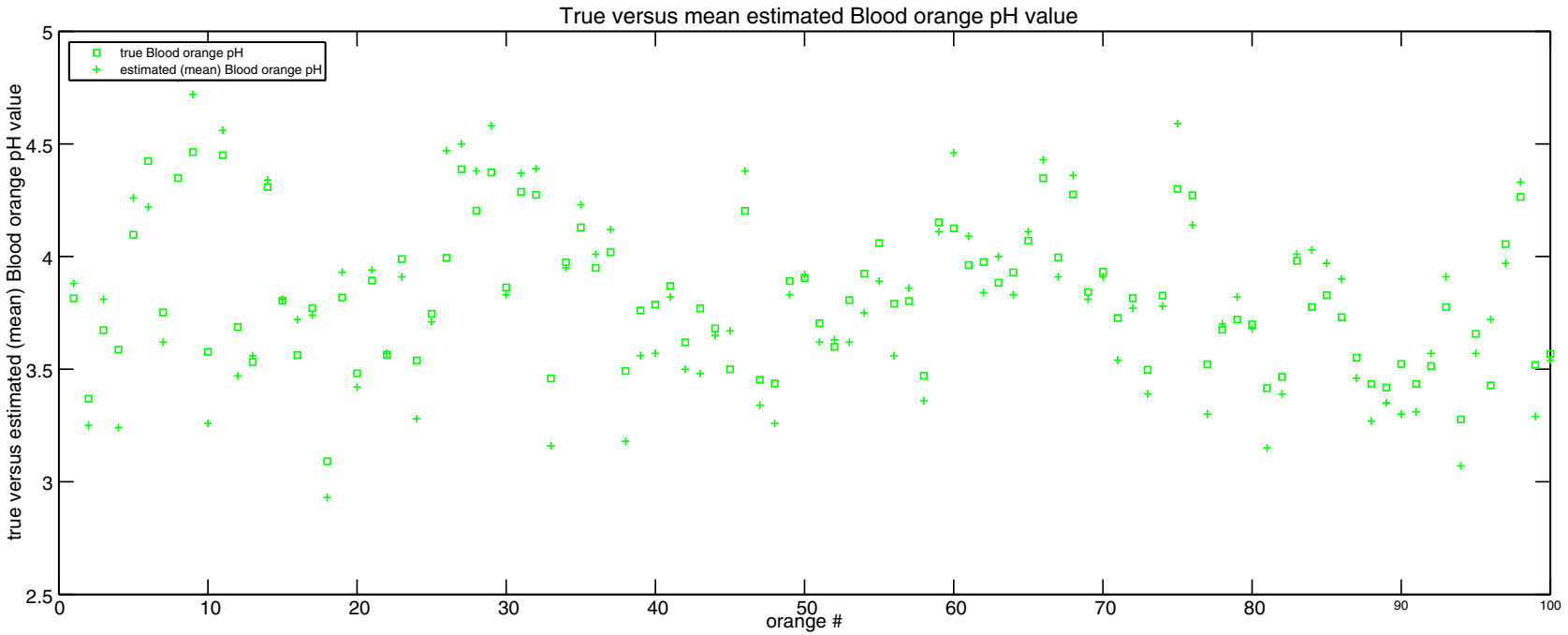


Fig. 9. True (green square) versus mean estimated Blood (green plus sign) orange pH values (test set): 1000 uniform random test set samples simulations averaged; each orange # is used 300 times inside the test set, on average. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

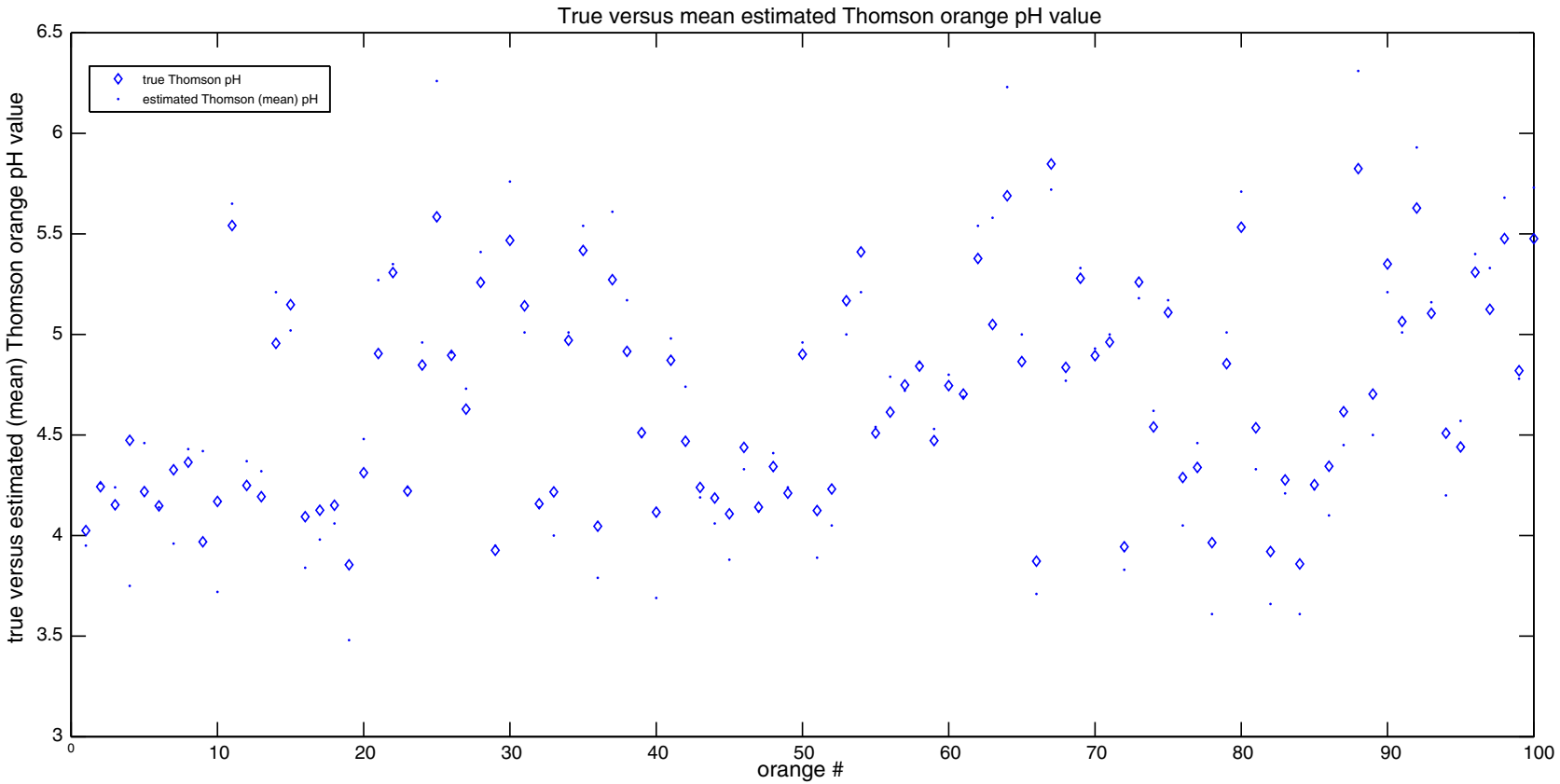


Fig. 10. True (blue diamond) versus mean estimated (blue dot) Thomson orange pH values (test set): 1000 uniform random test set samples simulations averaged; each orange # is used 300 times inside the test set, on average. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 10
True and mean ± std. estimated Bam orange pH values over the test set. Standard deviation of estimated pH values are also included, for the test set. A total of 1000 uniform random test set samples simulations averaged; on average, each orange # is used 300 times inside the test set, on average.

Orange#	True pH	Estimated pH	Orange#	True pH	Estimated pH	Orange#	True pH	Estimated pH
1	3.19	3.26 ± 0.15	*35*	4.47	4.46 ± 0.22	*69*	4.30	4.36 ± 0.18
2	3.43	3.36 ± 0.19	*36*	3.06	3.17 ± 0.14	*70*	3.98	3.87 ± 0.26
3	3.42	3.39 ± 0.14	*37*	4.52	4.22 ± 0.25	*71*	4.03	3.95 ± 0.21
4	3.03	3.67 ± 0.11	*38*	4.17	4.07 ± 0.17	*72*	3.08	3.18 ± 0.16
5	3.59	3.54 ± 0.19	*39*	3.62	3.57 ± 0.13	*73*	4.18	4.02 ± 0.18
6	3.34	3.23 ± 0.12	*40*	2.97	3.15 ± 0.21	*74*	3.72	3.87 ± 0.16
7	3.19	3.67 ± 0.13	*41*	4.02	4.12 ± 0.15	*75*	4.17	4.14 ± 0.15
8	3.57	3.67 ± 0.22	*42*	3.82	4.14 ± 0.15	*76*	3.26	3.48 ± 0.16
9	3.56	3.36 ± 0.29	*43*	3.38	3.62 ± 0.17	*77*	3.60	3.46 ± 0.13
10	3.00	3.20 ± 0.15	*44*	3.28	3.51 ± 0.13	*78*	2.91	3.06 ± 0.19
11	4.56	4.47 ± 0.18	*45*	3.13	3.34 ± 0.21	*79*	4.04	3.91 ± 0.14
12	3.52	3.34 ± 0.12	*46*	3.49	3.49 ± 0.15	*80*	4.60	4.57 ± 0.22
13	3.48	3.53 ± 0.25	*47*	3.32	3.29 ± 0.13	*81*	3.49	3.72 ± 0.19
14	4.19	4.13 ± 0.18	*48*	3.56	3.58 ± 0.18	*82*	2.95	3.18 ± 0.13
15	4.05	4.09 ± 0.18	*49*	3.42	3.52 ± 0.18	*83*	3.39	3.54 ± 0.14
16	3.09	3.30 ± 0.27	*50*	3.99	3.90 ± 0.15	*84*	2.91	3.16 ± 0.12
17	3.21	3.29 ± 0.14	*51*	3.13	3.21 ± 0.13	*85*	3.41	3.45 ± 0.15
18	3.27	3.29 ± 0.16	*52*	3.26	3.37 ± 0.13	*86*	3.31	3.45 ± 0.18
19	2.81	3.13 ± 0.22	*53*	4.03	4.22 ± 0.16	*87*	3.58	3.61 ± 0.16
20	3.61	3.43 ± 0.13	*54*	4.20	4.19 ± 0.20	*88*	4.89	4.72 ± 0.21
21	4.25	3.86 ± 0.15	*55*	3.66	3.68 ± 0.15	*89*	3.63	3.89 ± 0.13
22	4.32	4.21 ± 0.26	*56*	3.86	3.67 ± 0.15	*90*	4.20	4.05 ± 0.25
23	3.42	3.52 ± 0.13	*57*	3.81	3.93 ± 0.71	*91*	4.04	3.97 ± 0.20
24	3.99	3.98 ± 0.20	*58*	3.92	3.88 ± 0.21	*92*	4.78	4.30 ± 0.22
25	4.76	4.53 ± 0.14	*59*	3.66	3.68 ± 0.13	*93*	4.16	4.13 ± 0.23
26	3.96	3.87 ± 0.15	*60*	3.87	3.98 ± 0.21	*94*	3.39	3.53 ± 0.16
27	3.82	3.83 ± 0.15	*61*	3.78	3.84 ± 0.12	*95*	3.68	3.50 ± 0.15
28	4.36	4.11 ± 0.15	*62*	4.47	4.46 ± 0.17	*96*	4.35	4.18 ± 0.32
29	3.19	3.12 ± 0.22	*63*	4.50	4.15 ± 0.14	*97*	4.29	4.25 ± 0.15
30	4.64	4.28 ± 0.24	*64*	4.86	4.63 ± 0.16	*98*	4.58	4.52 ± 0.17
31	4.04	4.09 ± 0.15	*65*	4.03	3.92 ± 0.13	*99*	3.85	3.67 ± 0.14
32	3.34	3.39 ± 0.16	*66*	2.99	3.06 ± 0.26	*100*	4.62	4.37 ± 0.20
33	3.23	3.24 ± 0.16	*67*	4.61	4.66 ± 0.20			
34	4.04	4.04 ± 0.13	*68*	3.85	3.69 ± 0.15			

Table 11
True and mean ± std. estimated Blood orange pH values over the test set. Standard deviation of estimated pH values are also included, for the test set. A total of 1000 uniform random test set samples simulations averaged; on average, each orange # is used 300 times inside the test set, on average.

Orange#	True pH	Estimated pH	Orange#	True pH	Estimated pH	Orange#	True pH	Estimated pH
1	3.88	3.81 ± 0.12	*35*	4.23	4.13 ± 0.15	*69*	3.81	3.84 ± 0.19
2	3.25	3.37 ± 0.22	*36*	4.01	3.95 ± 0.13	*70*	3.91	3.93 ± 0.12
3	3.81	3.67 ± 0.11	*37*	4.12	4.02 ± 0.16	*71*	3.54	3.73 ± 0.18
4	3.24	3.59 ± 0.13	*38*	3.18	3.49 ± 0.13	*72*	3.77	3.81 ± 0.16
5	4.26	4.10 ± 0.17	*39*	3.56	3.76 ± 0.12	*73*	3.39	3.50 ± 0.17
6	4.22	4.42 ± 0.20	*40*	3.57	3.79 ± 0.15	*74*	3.78	3.83 ± 0.25
7	3.62	3.75 ± 0.11	*41*	3.82	3.87 ± 0.12	*75*	4.59	4.30 ± 0.26
8	4.79	4.35 ± 0.27	*42*	3.50	3.62 ± 0.13	*76*	4.14	4.27 ± 0.19
9	4.72	4.46 ± 0.15	*43*	3.48	3.77 ± 0.14	*77*	3.30	3.52 ± 0.12
10	3.26	3.58 ± 0.15	*44*	3.65	3.68 ± 0.14	*78*	3.70	3.68 ± 0.13
11	4.56	4.45 ± 0.29	*45*	3.67	3.50 ± 0.13	*79*	3.82	3.72 ± 0.12
12	3.47	3.69 ± 0.14	*46*	4.38	4.20 ± 0.14	*80*	3.68	3.70 ± 0.11
13	3.56	3.53 ± 0.16	*47*	3.34	3.45 ± 0.17	*81*	3.15	3.42 ± 0.14
14	4.34	4.31 ± 0.15	*48*	3.26	3.44 ± 0.17	*82*	3.39	3.47 ± 0.13
15	3.81	3.80 ± 0.11	*49*	3.83	3.89 ± 0.13	*83*	4.01	3.98 ± 0.13
16	3.72	3.56 ± 0.12	*50*	3.92	3.90 ± 0.11	*84*	4.03	3.78 ± 0.13
17	3.74	3.77 ± 0.11	*51*	3.62	3.70 ± 0.13	*85*	3.97	3.83 ± 0.12
18	2.93	3.09 ± 0.18	*52*	3.63	3.60 ± 0.13	*86*	3.90	3.73 ± 0.22
19	3.93	3.82 ± 0.11	*53*	3.62	3.81 ± 0.12	*87*	3.46	3.55 ± 0.14
20	3.42	3.48 ± 0.20	*54*	3.75	3.92 ± 0.23	*88*	3.27	3.43 ± 0.14
21	3.94	3.89 ± 0.14	*55*	3.89	4.06 ± 0.13	*89*	3.35	3.42 ± 0.14
22	3.57	3.56 ± 0.14	*56*	3.56	3.79 ± 0.14	*90*	3.30	3.52 ± 0.12
23	3.91	3.99 ± 0.12	*57*	3.86	3.80 ± 0.11	*91*	3.31	3.43 ± 0.12
24	3.28	3.54 ± 0.13	*58*	3.36	3.47 ± 0.16	*92*	3.57	3.51 ± 0.14
25	3.71	3.75 ± 0.13	*59*	4.11	4.15 ± 0.22	*93*	3.91	3.78 ± 0.15
26	4.47	3.99 ± 0.29	*60*	4.46	4.13 ± 0.21	*94*	3.07	3.28 ± 0.16
27	4.50	4.39 ± 0.14	*61*	4.09	3.96 ± 0.17	*95*	3.57	3.66 ± 0.15
28	4.38	4.20 ± 0.14	*62*	3.84	3.98 ± 0.12	*96*	3.72	3.43 ± 0.12
29	4.58	4.37 ± 0.13	*63*	4.00	3.88 ± 0.12	*97*	3.97	4.06 ± 0.18
30	3.83	3.86 ± 0.14	*64*	3.83	3.93 ± 0.12	*98*	4.33	4.26 ± 0.21
31	4.37	4.29 ± 0.15	*65*	4.11	4.07 ± 0.13	*99*	3.29	3.52 ± 0.13
32	4.39	4.27 ± 0.13	*66*	4.43	4.35 ± 0.15	*100*	3.54	3.60 ± 0.10
33	3.16	3.46 ± 0.24	*67*	3.91	4.00 ± 0.34			
34	3.95	3.97 ± 0.12	*68*	4.36	4.28 ± 0.13			

Table 12
True and mean ± std. estimated Thomson orange pH values over the test set. Standard deviation of estimated pH values are also included, for the test set. A total of 1000 uniform random test set samples simulations averaged; on average, each orange # is used 300 times inside the test set, on average.

Orange#	True pH	Estimated pH	Orange#	True pH	Estimated pH	Orange#	True pH	Estimated pH
1	3.95	4.02 ± 0.25	*35*	5.54	5.42 ± 0.29	*69*	5.33	5.28 ± 0.27
2	4.26	4.24 ± 0.21	*36*	3.79	4.05 ± 0.21	*70*	4.93	4.89 ± 0.27
3	4.24	4.15 ± 0.23	*37*	5.61	5.27 ± 0.35	*71*	5.00	4.96 ± 0.26
4	3.75	4.47 ± 0.18	*38*	5.17	4.92 ± 0.38	*72*	3.83	3.94 ± 0.24
5	4.46	4.22 ± 0.25	*39*	4.49	4.51 ± 0.29	*73*	5.18	5.26 ± 0.31
6	4.14	4.15 ± 0.23	*40*	3.69	4.12 ± 0.36	*74*	4.62	4.54 ± 0.22
7	3.96	4.33 ± 0.31	*41*	4.98	4.87 ± 0.27	*75*	5.17	5.11 ± 0.27
8	4.43	4.36 ± 0.25	*42*	4.74	4.47 ± 0.29	*76*	4.05	4.29 ± 0.25
9	4.42	3.97 ± 0.22	*43*	4.19	4.24 ± 0.23	*77*	4.46	4.34 ± 0.22
10	3.72	4.17 ± 0.27	*44*	4.06	4.19 ± 0.21	*78*	3.61	3.96 ± 0.25
11	5.65	5.54 ± 0.44	*45*	3.88	4.11 ± 0.23	*79*	5.01	4.85 ± 0.33
12	4.37	4.25 ± 0.23	*46*	4.33	4.44 ± 0.38	*80*	5.71	5.53 ± 0.28
13	4.32	4.19 ± 0.19	*47*	4.12	4.14 ± 0.21	*81*	4.33	4.54 ± 0.19
14	5.21	4.96 ± 0.20	*48*	4.41	4.34 ± 0.23	*82*	3.66	3.92 ± 0.25
15	5.02	5.15 ± 0.23	*49*	4.24	4.21 ± 0.22	*83*	4.21	4.28 ± 0.20
16	3.84	4.09 ± 0.28	*50*	4.96	4.90 ± 0.20	*84*	3.61	3.86 ± 0.23
17	3.98	4.13 ± 0.29	*51*	3.89	4.12 ± 0.28	*85*	4.23	4.25 ± 0.16
18	4.06	4.15 ± 0.21	*52*	4.05	4.23 ± 0.19	*86*	4.10	4.34 ± 0.24
19	3.48	3.85 ± 0.28	*53*	5.00	5.17 ± 0.24	*87*	4.45	4.62 ± 0.29
20	4.48	4.31 ± 0.19	*54*	5.21	5.41 ± 0.31	*88*	6.31	5.82 ± 0.29
21	5.27	4.91 ± 0.21	*55*	4.54	4.51 ± 0.24	*89*	4.50	4.70 ± 0.18
22	5.35	5.31 ± 0.32	*56*	4.79	4.61 ± 0.29	*90*	5.21	5.35 ± 0.26
23	4.24	4.22 ± 0.19	*57*	4.72	4.75 ± 0.19	*91*	5.01	5.06 ± 0.23
24	4.96	4.85 ± 0.19	*58*	4.86	4.84 ± 0.24	*92*	5.93	5.63 ± 0.38
25	6.26	5.58 ± 0.30	*59*	4.53	4.47 ± 0.30	*93*	5.16	5.10 ± 0.26
26	4.91	4.90 ± 0.22	*60*	4.80	4.75 ± 0.21	*94*	4.20	4.51 ± 0.23
27	4.73	4.63 ± 0.34	*61*	4.69	4.70 ± 0.17	*95*	4.57	4.44 ± 0.19
28	5.41	5.26 ± 0.32	*62*	5.54	5.38 ± 0.45	*96*	5.40	5.31 ± 0.25
29	3.95	3.93 ± 0.26	*63*	5.58	5.05 ± 0.20	*97*	5.33	5.12 ± 0.23
30	5.76	5.47 ± 0.32	*64*	6.23	5.69 ± 0.37	*98*	5.68	5.48 ± 0.25
31	5.01	5.14 ± 0.31	*65*	5.00	4.86 ± 0.22	*99*	4.78	4.82 ± 0.21
32	4.14	4.16 ± 0.19	*66*	3.71	3.87 ± 0.25	*100*	5.73	5.50 ± 0.30
33	4.00	4.22 ± 0.26	*67*	5.72	5.85 ± 0.33			
34	5.01	4.97 ± 0.22	*68*	4.77	4.84 ± 0.22			

using ANN-PSO for all three orange varieties (Bam, Blood and Thomson) as well as for the overall 3-variety 300 orange samples pH estimation system.

2.5. Neural network regressors: parameter optimization

MLP network has 9 scalar parameters listed next in 5 different groups: the number of layers, the number of neurons in each layer, the transfer function, back-propagation network training function, and back-propagation weight/bias learning function. In order to optimize MLP, all these parameters should be set at its optimal value. Table 8 shows all possible values for each parameter. Artificial bee colony (ABC) algorithm was used here for optimization of the 9 different parameters in MLP. In fact ABC algorithms consider these parameters as a vector. Optimization occurs by changing the value of each parameter in this vector and sending vector to the neural network to compute the output regression MSE error and record it at each time during training. The value of parameters in vector with less MSE is selected as the optimal value. Eq. (2) shows a set of design variables. This vector consists of nine variables that we want to optimize:

$$X = [x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9] \tag{2}$$

where x_1 is the number of inputs to the MLP network:

$$x_1 \epsilon \{1, 2, 3, \cdots, \max(input)\} \tag{3}$$

Variables x_2, x_3, and x_4 are the NN first and additional layers:

$$x_2 \epsilon \{1, 2, 3, \cdots, 25\} \tag{4}$$

$$x_3, x_4 \epsilon \{0, 1, 2, 3, \cdots 25\} \tag{5}$$

The range of the number of neurons in each layer is minimum 0 and maximum 25 neurons, {1,2,3, . . ., 25}, whereas for first layer it is between 1 and 25. In Eq. (6) variables x_5, x_6, and x_7 specify one out of possible 13 different transfer functions, for the first, second and third layers:

$$x_5, x_6, x_7 \epsilon \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13\} \tag{6}$$

All possible 13 transfer functions are listed in Table 8. Variable x_8 specifies the back-propagation network training function, one out of 19 possible different training functions:

$$x_8 \epsilon \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19\} \tag{7}$$

All possible 19 different back-propagation network training functions are listed in Table 8. Variable x_9 is the back-propagation weight/bias learning function:

$$x_9 \epsilon \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\} \tag{8}$$

A total of 15 different back-propagation weight/bias learning functions are listed in Table 8. Eq. (9) shows the final objective function:

$$Objective_function = minimize\{MSE(desired\ output, NN\ output)\} \tag{9}$$

Table 9 shows the optimal values of the neural network parameters that were determined using ABC algorithm and which will be used as such to get final optimal orange pH regression outputs results.

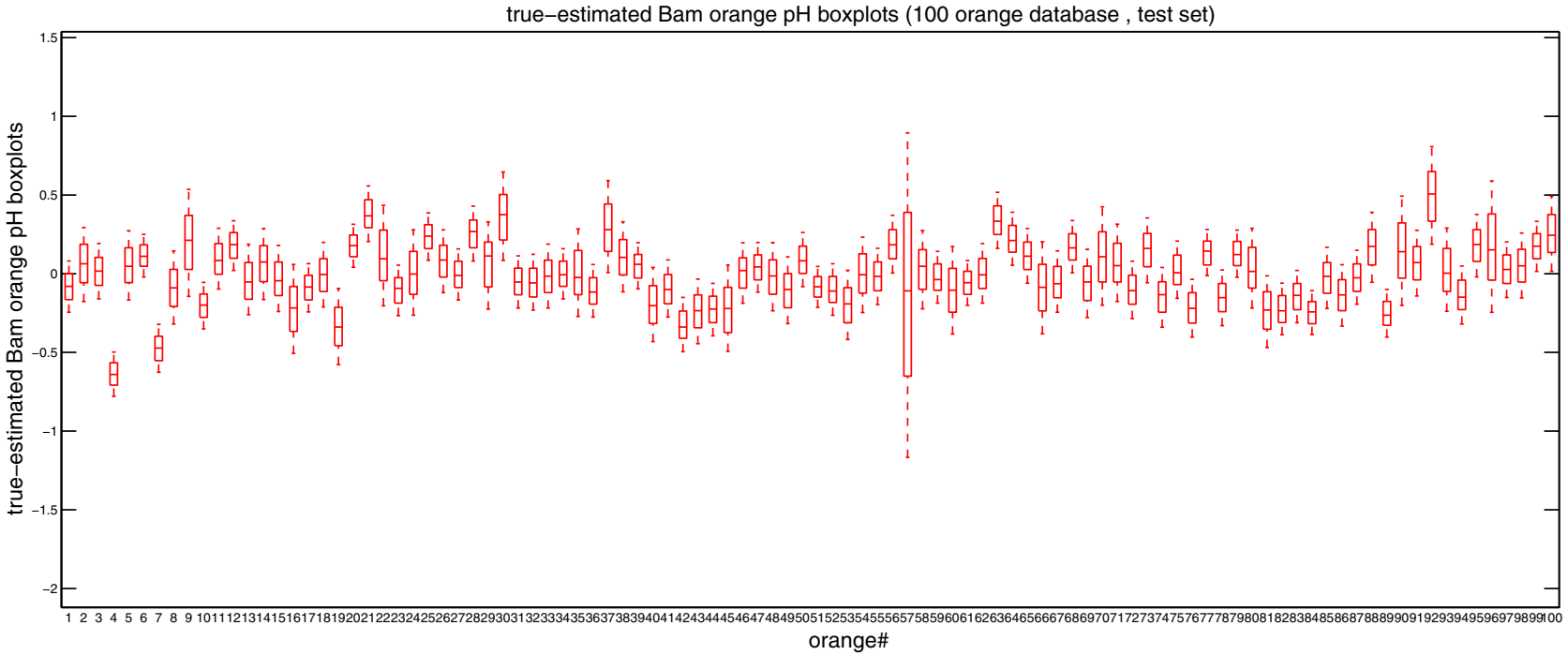


Fig. 11. True minus estimated Bam orange pH boxplot (test set): 1000 uniform random test set samples simulations averaged; each orange # is used 300 times inside the test set, on average.

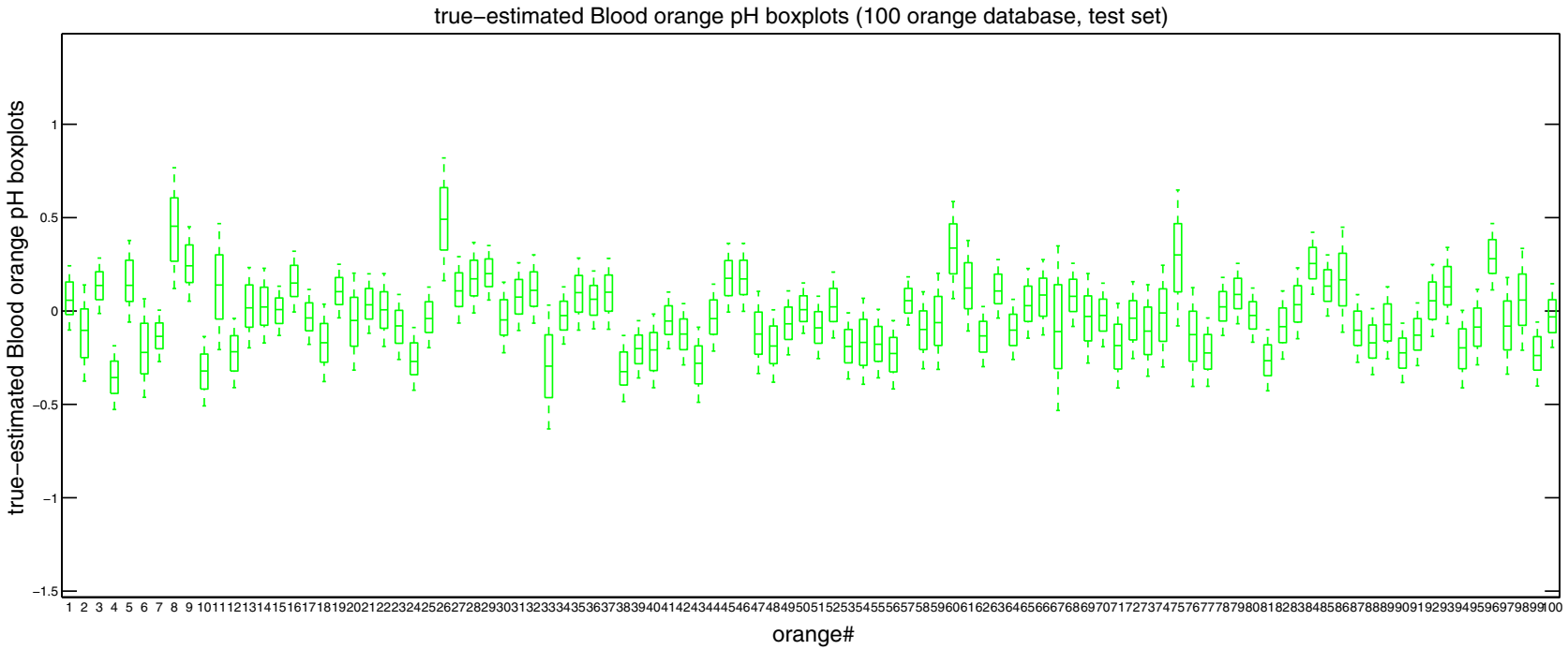


Fig. 12. True minus estimated Blood orange pH boxplot (test set): 1000 uniform random test set samples simulations averaged; each orange # is used 300 times inside the test set, on average.

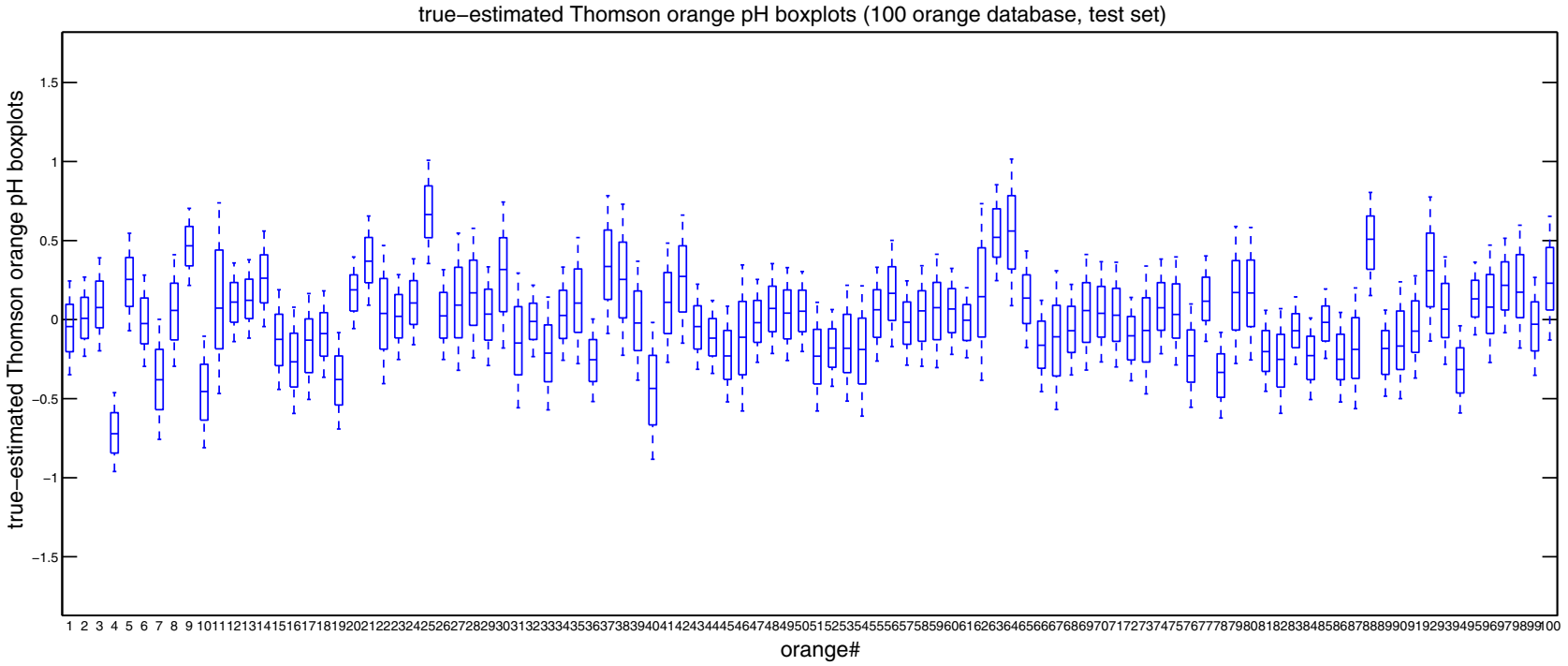


Fig. 13. True minus estimated Thomson orange pH boxplot (test set): 1000 uniform random test set samples simulations averaged; each orange # is used 300 times inside the test set, on average.

3. Results

In this section, we present a several statistical valid numerical (tabular) as well as visual results, including linear regression dispersion plots, true versus mean estimated pH value dispersion plots, true minus estimated pH value boxplots and several error coefficients, all over randomized test sets, for all 3 different orange varieties (Bam, Blood, Thomson), and for either 3 single-variety 100 orange samples each or for the 3-variety 300 orange samples, pH value regression problems. In all cases we used randomized trials to define train and test sample sets, using the following approach: 1000 uniform random test set samples simulations averaged, with 30% of total input data (30 random test samples for the single-variety 100 orange images system and 90 random test samples for the 3-variety 300 orange images system); each orange # is used $30 \times \frac{1000}{100} = 90 \times \frac{1000}{300} = 300$ times inside the test set, so each generated boxplot has 300 different pH values in it, on average.

3.1. Three single-variety pH regression problems

3.1.1. Linear regression plots

In Figs. 5–7, we plot the linear regression fit plots for Bam, Blood and Thomson orange varieties, respectively. The three orange varieties show a consistent similar performance.

3.1.2. True versus mean estimated pH values dispersion plots and tabular pH values

In Figs. 8–10, we plot true versus mean estimated orange pH values, for Bam, Blood and Thomson varieties, respectively. In addition, in Tables 10–12, we show the true and mean ± std. pH values, for Bam, Blood and Thomson orange varieties, respectively. Exception made of some minor visible differences, consistent results are shown over all three single orange varieties.

3.1.3. True minus estimated pH value boxplots

In Figs. 11–13, we plot true-estimated pH value boxplots, for Bam, Blood and Thomson orange varieties, respectively. All three varieties show a consistent performance.

3.1.4. Regression error coefficients

In Fig. 14, we plot MSE, RMSE and MAE boxplots for Bam (red), Blood (green) and Thomson (blue) orange varieties, all over the randomize test sets. At the same time, in Fig. 15 we plot regression coefficient R and the coefficient of determination R^2 boxplots for Bam (red), Blood (green) and Thomson (blue) orange varieties, all over the randomized trials test sets. As can be seen, all varieties show a consistent and very similar performance, with small visible differences shown.

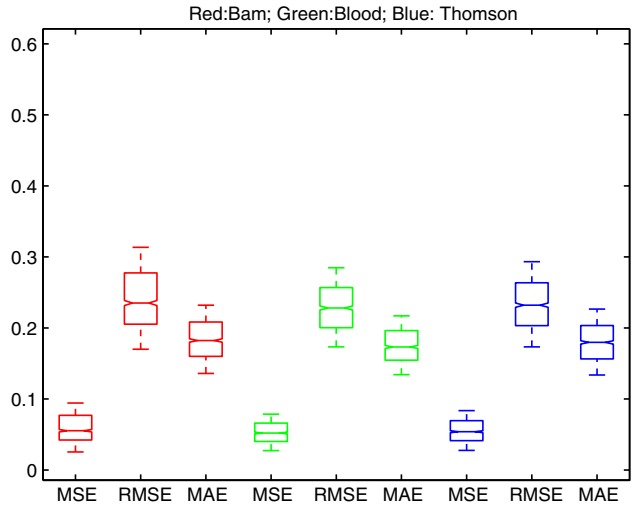


Fig. 14. Error indeces: MSE, RMSE and MAE boxplots computed for Bam (red), Blood (green) and Thomson (blue) orange varieties (test set): 1000 uniform random test set samples simulations averaged; each orange # is used 300 times inside the test set, on average. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

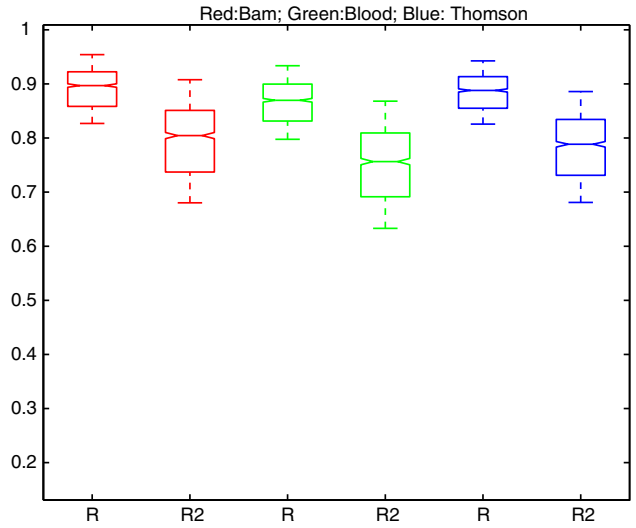


Fig. 15. Correlation indeces: R and R^2 boxplots computed for Bam (red), Blood (green) and Thomson (blue) orange varieties (test set): 1000 uniform random test set samples simulations averaged; each orange # is used 300 times inside the test set, on average. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.2. The 3-variety pH regression problem: three hundred Bam, Blood and Thomson oranges considered altogether

Next we show results for the 3-variety 300 orange samples pH regression problem, considering at the same time the 3 orange varieties altogether without distinction: Bam, Blood and Thomson samples. For obvious reasons, the performance of the 3-variety pH regression system shows a degradation in comparison with each of 3 single-variety counterparts, as shown next.

3.2.1. True versus mean estimated pH values dispersion plots and tabular pH values

In Fig. 16 we show the 3-variety 300 image orange samples true versus estimated orange pH values for the whole system (all 3 orange varieties together). In addition, in Table 13, we show the true and mean ± std. pH values for the 300 orange samples, all 3 varieties together, where first 100 samples belong to Bam variety, second 100 samples belong to Blood variety, and third 100 samples belong to Thomson orange variety. The accuracy of the 3-variety pH regression system is remarkable, particularly taking into account its automatic and non-intrusive nature.

3.2.2. True minus estimated pH value boxplots

Finally, in Fig. 17 we show the true minus estimated pH value boxplots for all 3 orange varieties altogether (at the same time), comprising a total of 300 input orange color image samples. The

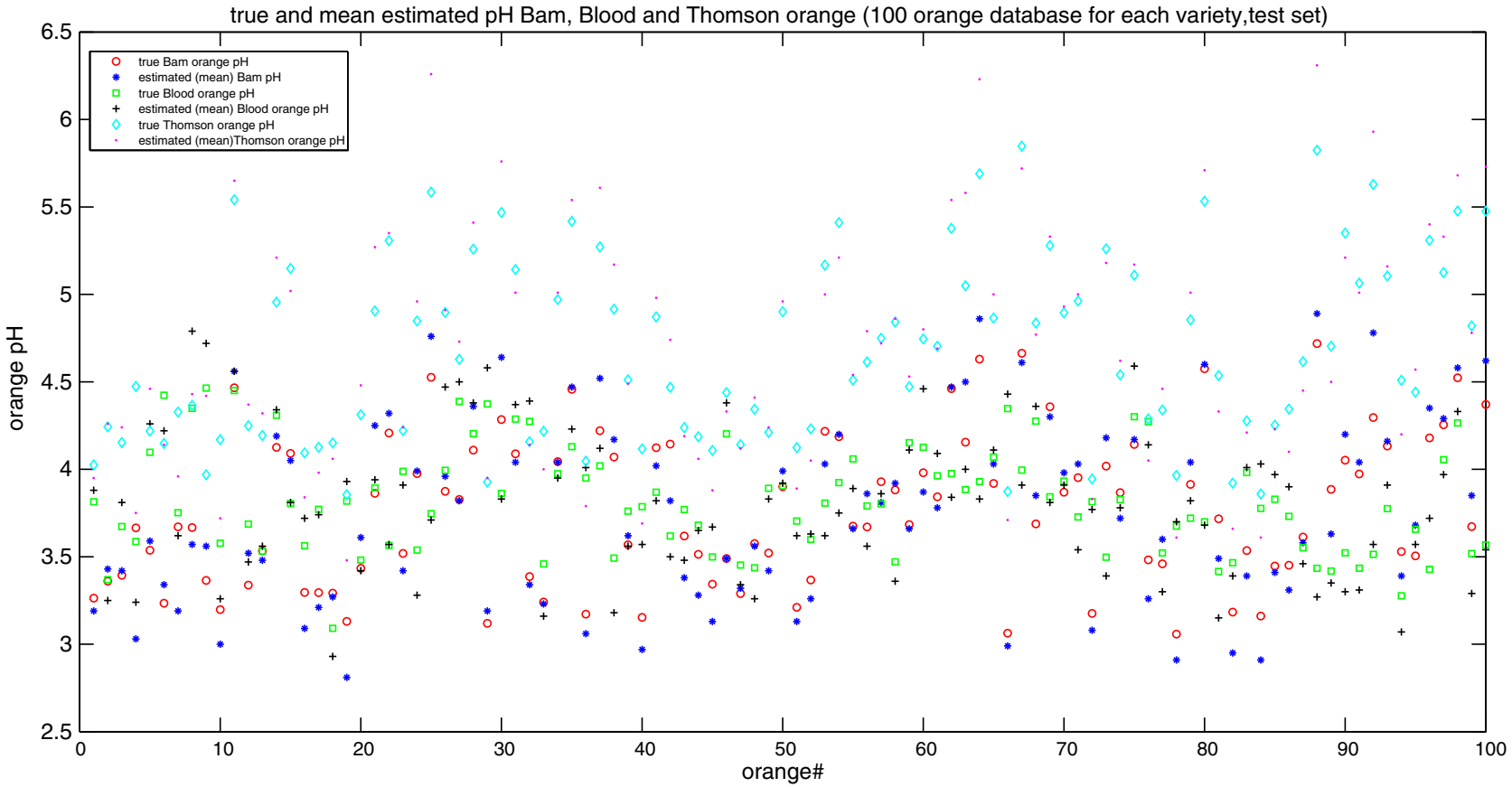


Fig. 16. True versus mean estimated 3-variety orange pH values (test set): 1000 uniform random test set samples simulations averaged; each orange # is used 300 times inside the test set, on average: true Bam pH (red circle), estimated mean Bam pH (red asterisk), true Blood pH (green square), estimated mean Blood (black plus sign), true Thomson (blue diamond) and estimated mean Thomson (purple dot). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 13
True versus mean ± std. estimated Bam (1–100), Blood (101–200) and Thomson (201–300) orange pH values, over the test set. A total of 1000 uniform random test set samples simulations averaged; on average, each orange # is used 300 times inside the test set, on average.

Orange#	True pH	Estimated pH	Orange#	True pH	Estimated pH	Orange#	True pH	Estimated pH
1	3.19	3.54 ± 0.17	*44*	3.28	3.81 ± 0.19	*87*	3.58	3.70 ± 0.15
2	3.43	3.64 ± 0.17	*45*	3.13	3.68 ± 0.21	*88*	4.89	4.40 ± 0.29
3	3.42	3.57 ± 0.19	*46*	3.49	3.71 ± 0.14	*89*	3.63	3.81 ± 0.17
4	3.03	3.75 ± 0.16	*47*	3.32	3.53 ± 0.16	*90*	4.20	4.04 ± 0.16
5	3.59	3.88 ± 0.21	*48*	3.56	3.71 ± 0.21	*91*	4.04	4.04 ± 0.18
6	3.34	3.51 ± 0.16	*49*	3.42	3.80 ± 0.20	*92*	4.78	4.17 ± 0.19
7	3.19	3.67 ± 0.17	*50*	3.99	3.91 ± 0.14	*93*	4.16	4.12 ± 0.19
8	3.57	3.75 ± 0.14	*51*	3.13	3.50 ± 0.15	*94*	3.39	3.65 ± 0.16
9	3.56	4.05 ± 0.40	*52*	3.26	3.56 ± 0.13	*95*	3.68	3.73 ± 0.15
10	3.00	3.47 ± 0.15	*53*	4.03	4.04 ± 0.16	*96*	4.35	4.23 ± 0.20
11	4.56	4.21 ± 0.25	*54*	4.20	4.08 ± 0.21	*97*	4.29	4.13 ± 0.17
12	3.52	3.53 ± 0.15	*55*	3.66	3.72 ± 0.15	*98*	4.58	4.27 ± 0.20
13	3.48	3.70 ± 0.15	*56*	3.86	3.73 ± 0.14	*99*	3.85	3.84 ± 0.17
14	4.19	3.99 ± 0.15	*57*	3.81	4.04 ± 0.27	*100*	4.62	4.20 ± 0.18
15	4.05	4.00 ± 0.15	*58*	3.92	3.93 ± 0.15	*101*	3.88	3.49 ± 0.16
16	3.09	3.68 ± 0.16	*59*	3.66	3.81 ± 0.15	*102*	3.25	3.67 ± 0.17
17	3.21	3.53 ± 0.14	*60*	3.87	3.88 ± 0.15	*103*	3.81	3.53 ± 0.18
18	3.27	3.67 ± 0.17	*61*	3.78	3.80 ± 0.14	*104*	3.24	3.72 ± 0.17
19	2.81	3.56 ± 0.22	*62*	4.47	4.25 ± 0.22	*105*	4.26	3.83 ± 0.21
20	3.61	3.60 ± 0.13	*63*	4.50	4.03 ± 0.20	*106*	4.22	3.45 ± 0.15
21	4.25	3.88 ± 0.16	*64*	4.86	4.32 ± 0.26	*107*	3.62	3.65 ± 0.18
22	4.32	4.30 ± 0.25	*65*	4.03	3.95 ± 0.16	*108*	4.79	3.74 ± 0.15
23	3.42	3.61 ± 0.14	*66*	2.99	3.65 ± 0.31	*109*	4.72	3.78 ± 0.38
24	3.99	3.85 ± 0.33	*67*	4.61	4.49 ± 0.24	*110*	3.26	3.46 ± 0.17
25	4.76	4.30 ± 0.25	*68*	3.85	3.86 ± 0.19	*111*	4.56	4.21 ± 0.24
26	3.96	3.91 ± 0.15	*69*	4.30	4.16 ± 0.21	*112*	3.47	3.55 ± 0.17
27	3.82	3.86 ± 0.14	*70*	3.98	3.84 ± 0.19	*113*	3.56	3.71 ± 0.17
28	4.36	4.04 ± 0.16	*71*	4.03	3.82 ± 0.22	*114*	4.34	3.99 ± 0.17
29	3.19	3.62 ± 0.21	*72*	3.08	3.53 ± 0.17	*115*	3.81	4.00 ± 0.15
30	4.64	4.09 ± 0.17	*73*	4.18	4.00 ± 0.28	*116*	3.72	3.65 ± 0.19
31	4.04	4.01 ± 0.14	*74*	3.72	3.86 ± 0.14	*117*	3.74	3.50 ± 0.14
32	3.34	3.55 ± 0.14	*75*	4.17	4.10 ± 0.20	*118*	2.93	3.66 ± 0.17
33	3.23	3.55 ± 0.20	*76*	3.26	3.77 ± 0.18	*119*	3.93	3.55 ± 0.23
34	4.04	3.91 ± 0.16	*77*	3.60	3.60 ± 0.15	*120*	3.42	3.62 ± 0.14
35	4.47	4.28 ± 0.21	*78*	2.91	3.50 ± 0.16	*121*	3.94	3.90 ± 0.18
36	3.06	3.48 ± 0.15	*79*	4.04	3.87 ± 0.13	*122*	3.57	4.36 ± 0.24
37	4.52	4.25 ± 0.26	*80*	4.60	4.43 ± 0.26	*123*	3.91	3.61 ± 0.14
38	4.17	3.97 ± 0.16	*81*	3.49	3.84 ± 0.16	*124*	3.28	4.02 ± 0.27
39	3.62	3.63 ± 0.14	*82*	2.95	3.46 ± 0.16	*125*	3.71	4.40 ± 0.22
40	2.97	3.64 ± 0.31	*83*	3.39	3.72 ± 0.15	*126*	4.47	3.90 ± 0.15
41	4.02	4.05 ± 0.17	*84*	2.91	3.52 ± 0.22	*127*	4.50	3.85 ± 0.15
42	3.82	4.05 ± 0.19	*85*	3.41	3.74 ± 0.15	*128*	4.38	4.05 ± 0.17
43	3.38	3.77 ± 0.23	*86*	3.31	3.69 ± 0.24	*129*	4.58	3.51 ± 0.21
130	3.83	4.14±0.18	*178*	3.70	3.43±0.17	*226*	4.91	4.52±0.26
131	4.37	4.01±0.16	*179*	3.82	3.89±0.15	*227*	4.73	4.55±0.48
132	4.39	3.51±0.14	*180*	3.68	4.55±0.25	*228*	5.41	4.35±0.23
133	3.16	3.54±0.19	*181*	3.15	3.84±0.16	*229*	3.95	4.57±0.24
134	3.95	3.93±0.15	*182*	3.39	3.44±0.17	*230*	5.76	4.48±0.23
135	4.23	4.27±0.23	*183*	4.01	3.69±0.16	*231*	5.01	4.43±0.26
136	4.01	3.42±0.15	*184*	4.03	3.39±0.23	*232*	4.14	4.56±0.24
137	4.12	4.29±0.26	*185*	3.97	3.72±0.15	*233*	4.00	4.57±0.24
138	3.18	3.99±0.15	*186*	3.90	3.61±0.24	*234*	5.01	4.60±0.26
139	3.56	3.64±0.12	*187*	3.46	3.71±0.15	*235*	5.54	4.44±0.22
140	3.57	3.52±0.35	*188*	3.27	4.58±0.26	*236*	3.79	4.58±0.24
141	3.82	4.07±0.20	*189*	3.35	3.82±0.15	*237*	5.61	4.45±0.21
142	3.50	4.06±0.19	*190*	3.30	4.09±0.17	*238*	5.17	4.53±0.25
143	3.48	3.77±0.23	*191*	3.31	4.08±0.17	*239*	4.49	4.52±0.26
144	3.65	3.77±0.19	*192*	3.57	4.26±0.20	*240*	3.69	4.54±0.34
145	3.67	3.62±0.23	*193*	3.91	4.14±0.17	*241*	4.98	4.59±0.36
146	4.38	3.69±0.15	*194*	3.07	3.65±0.16	*242*	4.74	4.41±0.32
147	3.34	3.51±0.16	*195*	3.57	3.75±0.18	*243*	4.19	4.47±0.26
148	3.26	3.72±0.20	*196*	3.72	4.26±0.20	*244*	4.06	4.53±0.23
149	3.83	3.75±0.21	*197*	3.97	4.15±0.18	*245*	3.88	4.65±0.27
150	3.92	3.93±0.16	*198*	4.33	4.29±0.22	*246*	4.33	4.28±0.34
151	3.62	3.49±0.16	*199*	3.29	3.86±0.17	*247*	4.12	4.49±0.30
152	3.63	3.56±0.14	*200*	3.54	4.24±0.18	*248*	4.41	4.47±0.26
153	3.62	4.06±0.16	*201*	3.95	4.77±0.36	*249*	4.24	4.50±0.28
154	3.75	4.11±0.21	*202*	4.26	4.56±0.29	*250*	4.96	4.60±0.25
155	3.89	3.70±0.14	*203*	4.24	4.50±0.26	*251*	3.89	4.33±0.33
156	3.56	3.75±0.15	*204*	3.75	4.18±0.26	*252*	4.05	4.67±0.31
157	3.86	4.03±0.29	*205*	4.46	4.21±0.30	*253*	5.00	4.38±0.29

Table 13 (*continued*)

Orange#	True pH	Estimated pH	Orange#	True pH	Estimated pH	Orange#	True pH	Estimated pH
158	3.36	3.96±0.16	*206*	4.14	3.71±0.19	*254*	5.21	4.52±0.24
159	4.11	3.80±0.15	*207*	3.96	4.63±0.27	*255*	4.54	4.74±0.37
160	4.46	3.87±0.15	*208*	4.43	4.55±0.26	*256*	4.79	4.54±0.28
161	4.09	3.79±0.14	*209*	4.42	4.49±0.24	*257*	4.72	4.53±0.23
162	3.84	4.31±0.22	*210*	3.72	4.28±0.19	*258*	4.86	4.29±0.45
163	4.00	4.06±0.19	*211*	5.65	4.46±0.29	*259*	4.53	4.69±0.30
164	3.83	4.38±0.23	*212*	4.37	4.50±0.29	*260*	4.80	4.55±0.24
165	4.11	3.93±0.15	*213*	4.32	4.52±0.24	*261*	4.69	4.52±0.26
166	4.43	3.39±0.23	*214*	5.21	4.41±0.25	*262*	5.54	4.22±0.34
167	3.91	4.54±0.24	*215*	5.02	4.61±0.27	*263*	5.58	4.63±0.28
168	4.36	3.80±0.19	*216*	3.84	4.54±0.38	*264*	6.23	4.27±0.27
169	3.81	4.21±0.21	*217*	3.98	4.79±0.39	*265*	5.00	4.46±0.41
170	3.91	3.86±0.21	*218*	4.06	4.54±0.27	*266*	3.71	4.47±0.28
171	3.54	3.90±0.20	*219*	3.48	4.58±0.29	*267*	5.72	4.20±0.30
172	3.77	3.48±0.17	*220*	4.48	4.61±0.44	*268*	4.77	4.64±0.34
173	3.39	4.08±0.24	*221*	5.27	4.53±0.27	*269*	5.33	4.21±0.23
174	3.78	3.87±0.15	*222*	5.35	4.75±0.46	*270*	4.93	4.21±0.19
175	4.59	4.07±0.19	*223*	4.24	4.58±0.28	*271*	5.00	4.20±0.17
176	4.14	3.70±0.18	*224*	4.96	4.34±0.34	*272*	3.83	4.67±0.25
177	3.30	3.62±0.14	*225*	6.26	4.55±0.33	*273*	5.18	4.33±0.49
274	4.62	4.44±0.57	*283*	4.21	4.64±0.26	*292*	5.93	4.67±0.42
275	5.17	4.38±0.32	*284*	3.61	4.52±0.28	*293*	5.16	4.27±0.33
276	4.05	4.50±0.26	*285*	4.23	4.57±0.26	*294*	4.20	4.20±0.31
277	4.46	4.17±0.24	*286*	4.10	3.65±0.16	*295*	4.57	4.51±0.31
278	3.61	4.76±0.34	*287*	4.45	4.50±0.44	*296*	5.40	4.49±0.22
279	5.01	4.92±0.59	*288*	6.31	4.66±0.34	*297*	5.33	4.55±0.22
280	5.71	4.58±0.26	*289*	4.50	4.50±0.25	*298*	5.68	4.65±0.27
281	4.33	4.67±0.28	*290*	5.21	4.55±0.30	*299*	4.78	4.50±0.26
282	3.66	4.46±0.31	*291*	5.01	4.26±0.22	*300*	5.73	4.47±0.24

accuracy of this last system is remarkable, despite is bellow single-variety counterparts for obvious reasons.

4. Conclusion

The novelty of this work is emphasized next:

1. Non-intrusive estimation of pH values in orange (Bam, Blood, Thomson) was performed based on a cheap computer vision method. Non-destructive estimation of the pH properties of oranges is an important goal in the food industry, since online operations, such as fruit packaging based on the amount of pH that can determine the degree of acidity and alkalinity of oranges and infer the different stages of handling, based on this estimation.
2. The system could be potentially adapted and used in a real industrial environment with color non-high resolution images, given its proven accuracy.
3. The optimal adjustment of the different parameters of MLP which include: the number of layers, the number of neurons in each layer, the transfer function, back-propagation network training function, and back-propagation weight/bias learning function during training stage, based on an artificial bee colony algorithm.

Accurate and automatic estimation of pH value using a non-destructive method might be useful, since it would allow the proper grading of fruit based on pH. For this reason, in this study a method for measuring pH in three different orange varieties (Bam, Blood and Thomson) is proposed, as summarized in Fig. 18. Main results are summarized next to conclude:

1. Among all extracted features, only color and texture features based on the gray level co-occurrence matrix were used in the non-destructive prediction of pH values in orange fruit.
2. Given that the various orange variety peels are rather different from each other, it was not easy to build a multi-variety automatic pH estimation system from orange peel color image, despite remarkable results are shown.
3. The average time used for estimating pH based on the here proposed non-destructive imaging method in orange was 0.42 s total, on average, from which 0.15 s took image segmentation, 0.13 s took feature extraction and selection and 0.14 s took MLP neural network pH value regression, on average. Those time values should be good for an online industrial application and could be reduced further using a proper compiled programming language.
4. Results show that the proposed method was able to predict the pH value from external visible-range peel image with high accuracy over three different orange varieties either as 3 single-variety systems or even as one 3-variety single system, although a degradation of accuracy is observed in the last case for obvious reasons.



Fig. 17. True minus estimated 3-variety orange pH boxplot (test set): 1000 uniform random test set samples simulations averaged; each orange # is used 300 times inside the test set, on average: Bam (red), Blood (green) and Thomson (blue). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

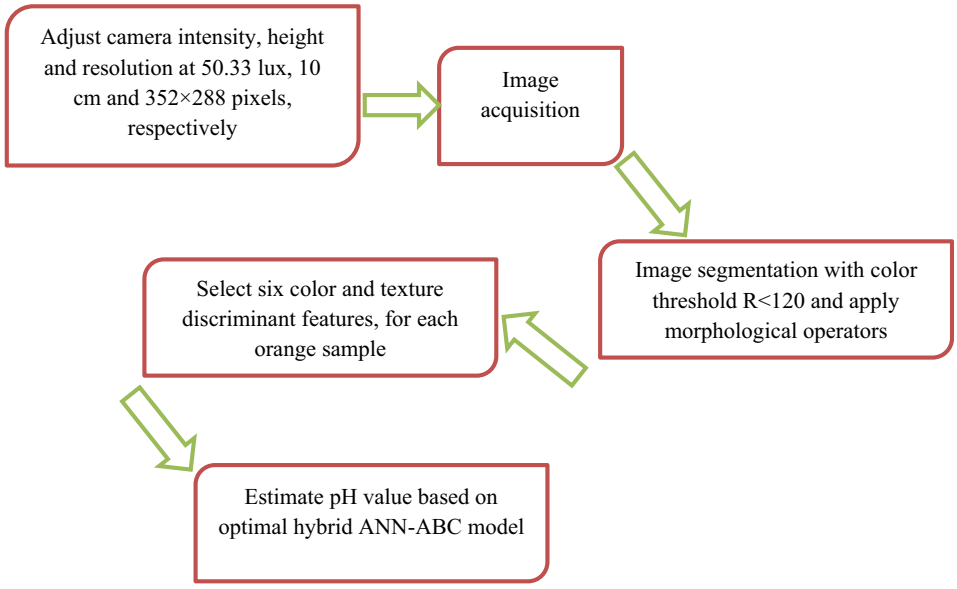


Fig. 18. Final computer vision system designed to estimate orange pH non-intrusively.

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Appendix A. Supplementary data

As supplementary material, we present here the 3-variety 300 orange visible-range color image database for the three orange classes, 100 images each, which might be freely available to researchers on properly justified requests, as follows: Bam orange variety (Fig. S1), Blood orange variety (Fig. S2) and Thomson orange variety (Fig. S3). For reproducibility purposes, we also provide the Matlab .fig result figures, for potential comparison purposes in the future. Supplementary data to this article can be found online at https://doi.org/10.1016/j.measurement.2019.107298.

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